

CSCI322

Systems Administration

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Network management

- Keep the network healthy
 - Fault detection for networks, gateways, and critical servers
 - Schemes for notifying an administrator of problems
 - General network monitoring
 - Documentation and visualisation of the network
 - Administration of network devices from a central site



FCAPS

- **F**ault Management
- **C**onfiguration Management
- **A**ccounting Management
- **P**erformance Management
- **S**ecurity Management



Some principles first

- Make one change at a time.
- Document the situation as it was before you got involved, and document every change you make
- Start at one end of a network and work through the system's critical component
- Communicate regularly
- Work as a team
- Use the layers of the network to negotiate the problem. Start at the “top” or “bottom”.



Network trouble shooting

- Do you have physical connectivity?
- Is your interface configured properly?
- Do your ARP tables show other hosts?
- Is there a firewall on your local machine?
- Is there a firewall anywhere between you and your destination?
- If firewalls are involved, do they pass ping packets?
- Can you ping the localhost?
- Can you ping other local hosts by IP address?
- Is DNS working properly?
- Can you ping other local hosts by hostname?
- Can you ping hosts on another network?
- Do high level services work?



ping

- The *ping* command send an ICMP ECHO_REQUEST packet to a target host and waits to see if the host answers back.
- Routing tables, physical networks, and gateways are all involved.
- The output for ping shows the host's IP address, the ICMP sequence number of each response packet, and the round trip travel time.
- Use `-n` option if you suspect that DNS is not working



What information can you get from ping

- ICMP sequence number tells you if packets have been dropped. If packets have been dropped, then
 - run *tracert* to discover the route to the target host
 - *ping* the intermediate gateways in sequence
- The round trip time gives you insights into the overall performance of a path through a network.



traceroute

- The traceroute command uncovers the sequence of gateways through which an IP packet travels to reach its destination.
- traceroute works by setting the time-to-live field (TTL) of an outbound packet to an artificially low number.
- There are traceroute-like utilities, try *mtr*
- Inverse traceroute, see <http://traceroute.org>

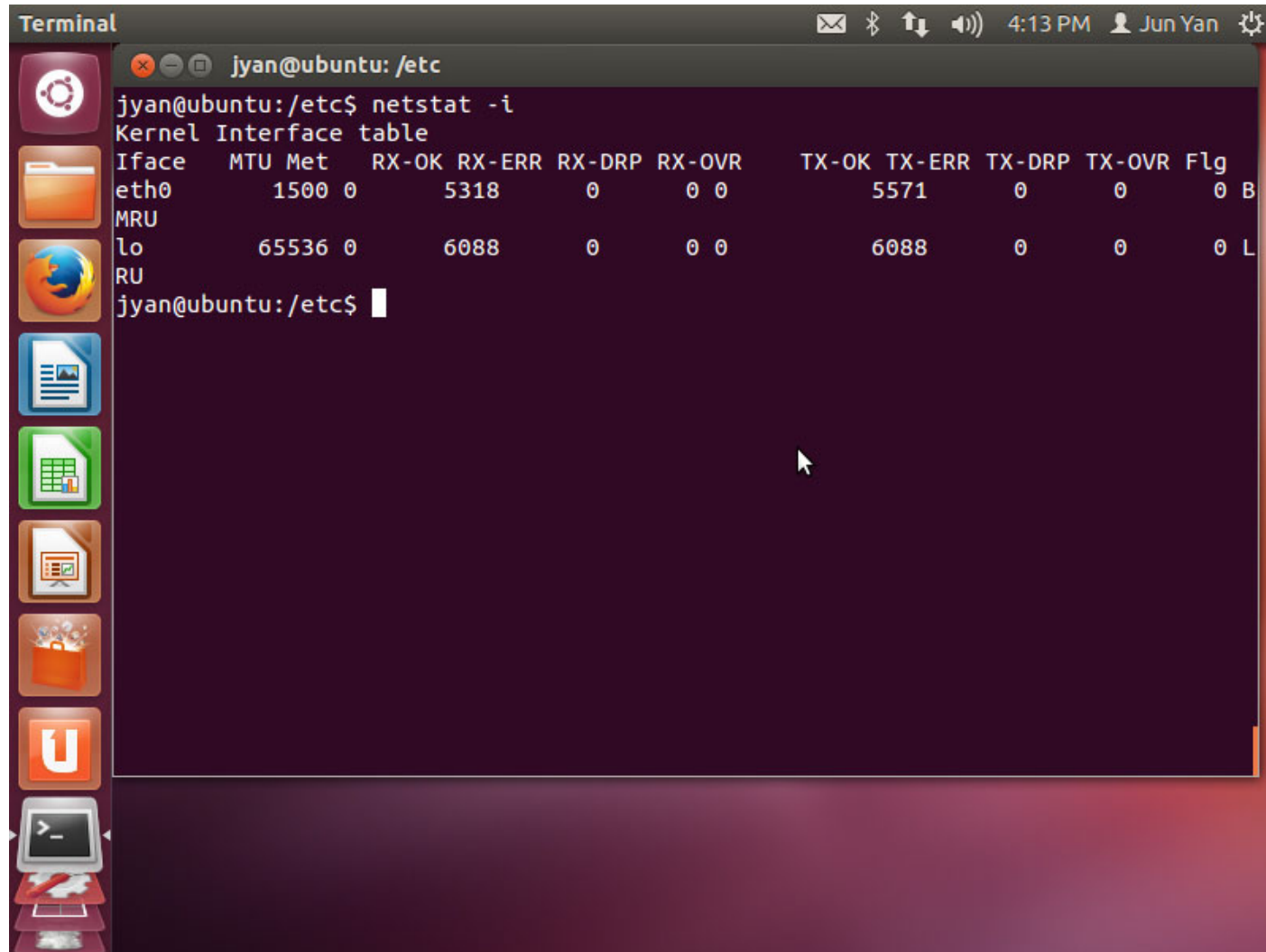


netstat

- *netstat* collects a wealth of information about the state of your computer's networking software
 - Interface statistics
 - Routing information
 - Connection tables
- *netstat* can be used to
 - Inspect interface configuration information
 - Monitor the status of network connections
 - Identify listening network services
 - Examine the routing table
 - View the operational statistics for various network protocols



Inspecting interface configuration



A terminal window titled "Terminal" showing the output of the command `netstat -i`. The window has a dark purple background and a light-colored terminal text. The output displays the Kernel Interface table with columns for interface name, MTU, memory, and various statistics. The interfaces shown are `eth0` and `lo`. The `lo` interface has a large MTU of 65536. The `eth0` interface has a MTU of 1500. The statistics for `eth0` show 5318 RX-OK, 0 RX-ERR, 0 RX-DRP, 0 RX-OVR, 5571 TX-OK, 0 TX-ERR, 0 TX-DRP, and 0 TX-OVR. The `lo` interface has 6088 RX-OK, 0 RX-ERR, 0 RX-DRP, 0 RX-OVR, 6088 TX-OK, 0 TX-ERR, 0 TX-DRP, and 0 TX-OVR. The window also shows the user `jyan@ubuntu: /etc` and the time `4:13 PM`.

```
jyan@ubuntu: /etc$ netstat -i
Kernel Interface table
Iface  MTU Met  RX-OK RX-ERR RX-DRP  RX-OVR    TX-OK TX-ERR TX-DRP  TX-OVR Flg
eth0    1500 0      5318   0      0 0        5571    0      0      0 0 B
MRU
lo      65536 0      6088   0      0 0        6088    0      0      0 0 L
RU
jyan@ubuntu: /etc$
```



```
Terminal
jyan@ubuntu: /etc
lo      65536 0      6088      0      0 0      6088      0      0      0 L
RU
jyan@ubuntu:/etc$ ifconfig -a
eth0      Link encap:Ethernet  HWaddr 00:0c:29:fe:77:5c
          inet addr:192.168.80.135  Bcast:192.168.80.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe77:5c/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:5324 errors:0 dropped:0 overruns:0 frame:0
          TX packets:5577 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:709985 (709.9 KB)  TX bytes:506208 (506.2 KB)
          Interrupt:19 Base address:0x2000

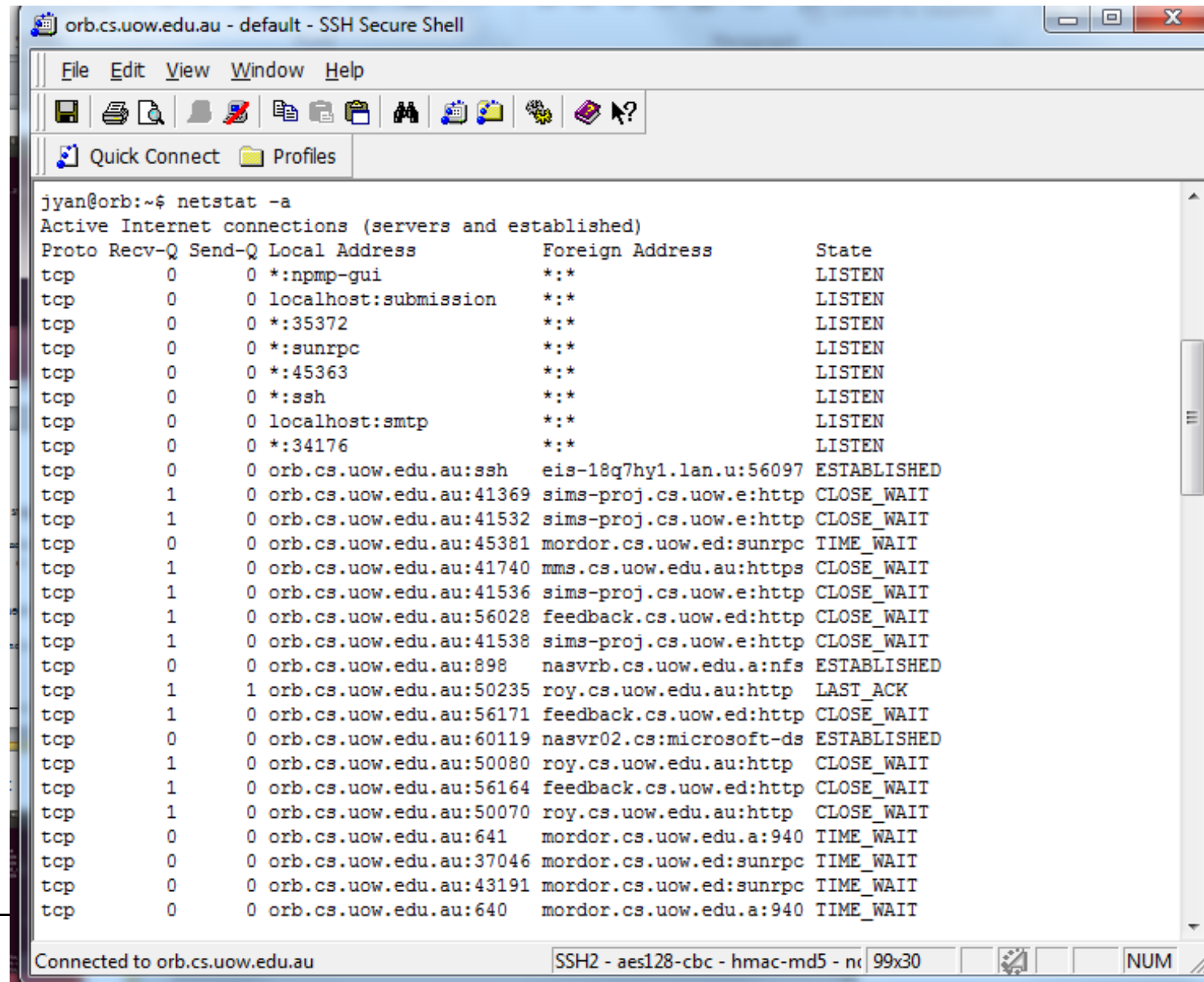
lo      Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:6096 errors:0 dropped:0 overruns:0 frame:0
          TX packets:6096 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:567709 (567.7 KB)  TX bytes:567709 (567.7 KB)

jyan@ubuntu:/etc$
```

- Errors can show up. A large number of errors on a single machine suggests a problem
 - What are the error rates of the neighbouring machines?
 - Is there a problem with that machine's interface or connection?
 - Or is there a media or network problem?
- You should not see collisions even when the network is under heavy load
 - Is the flow control enabled on your switches and routers?



Monitoring the status of network



```
orb.cs.uow.edu.au - default - SSH Secure Shell
File Edit View Window Help
Quick Connect Profiles
jyan@orb:~$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 *:nmp-gui               *:                       LISTEN
tcp        0      0 localhost:submission    *:                       LISTEN
tcp        0      0 *:35372                 *:                       LISTEN
tcp        0      0 *:sunrpc                *:                       LISTEN
tcp        0      0 *:45363                 *:                       LISTEN
tcp        0      0 *:ssh                   *:                       LISTEN
tcp        0      0 localhost:smtp          *:                       LISTEN
tcp        0      0 *:34176                 *:                       LISTEN
tcp        0      0 orb.cs.uow.edu.au:ssh   eis-18q7hyl.lan.u:56097 ESTABLISHED
tcp        1      0 orb.cs.uow.edu.au:41369 sims-proj.cs.uow.e:http CLOSE_WAIT
tcp        1      0 orb.cs.uow.edu.au:41532 sims-proj.cs.uow.e:http CLOSE_WAIT
tcp        0      0 orb.cs.uow.edu.au:45381 mordor.cs.uow.ed:sunrpc TIME_WAIT
tcp        1      0 orb.cs.uow.edu.au:41740 mms.cs.uow.edu.au:https CLOSE_WAIT
tcp        1      0 orb.cs.uow.edu.au:41536 sims-proj.cs.uow.e:http CLOSE_WAIT
tcp        1      0 orb.cs.uow.edu.au:56028 feedback.cs.uow.ed:http CLOSE_WAIT
tcp        1      0 orb.cs.uow.edu.au:41538 sims-proj.cs.uow.e:http CLOSE_WAIT
tcp        0      0 orb.cs.uow.edu.au:898   nasvrb.cs.uow.edu.a:nfs ESTABLISHED
tcp        1      1 orb.cs.uow.edu.au:50235 roy.cs.uow.edu.au:http  LAST_ACK
tcp        1      0 orb.cs.uow.edu.au:56171 feedback.cs.uow.ed:http CLOSE_WAIT
tcp        0      0 orb.cs.uow.edu.au:60119 nasvr02.cs:microsoft-ds ESTABLISHED
tcp        1      0 orb.cs.uow.edu.au:50080 roy.cs.uow.edu.au:http  CLOSE_WAIT
tcp        1      0 orb.cs.uow.edu.au:56164 feedback.cs.uow.ed:http CLOSE_WAIT
tcp        1      0 orb.cs.uow.edu.au:50070 roy.cs.uow.edu.au:http  CLOSE_WAIT
tcp        0      0 orb.cs.uow.edu.au:641   mordor.cs.uow.edu.a:940 TIME_WAIT
tcp        0      0 orb.cs.uow.edu.au:37046 mordor.cs.uow.ed:sunrpc TIME_WAIT
tcp        0      0 orb.cs.uow.edu.au:43191 mordor.cs.uow.ed:sunrpc TIME_WAIT
tcp        0      0 orb.cs.uow.edu.au:640   mordor.cs.uow.edu.a:940 TIME_WAIT
Connected to orb.cs.uow.edu.au  SSH2 - aes128-cbc - hmac-md5 - nc 99x30  NUM
```

- Send-Q and Recv-Q show the sizes of the local host's send and receive queues for the connection
 - These numbers should tend toward 0
- State has meaning for TCP only
 - LISTEN
 - ESTABLISHED
 - TIME_WAIT
- netstat -a is primarily useful for debugging higher-level problems once you have determined that basic networking facilities are working correctly
 - E.g., the state SYN_SENT identifies a process that is trying to contact a non-existent network server
 - What about SYN_WAIT?



Identifying listening network services

```
orb.cs.uow.edu.au - default - SSH Secure Shell
File Edit View Window Help
Quick Connect Profiles
jyan@orb:~$ clear
jyan@orb:~$ netstat -lp
(No info could be read for "-p": geteuid()=12570 but you should be root.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 *:npmp-gui              *:*                     LISTEN      -
tcp        0      0 localhost:submission    *:*                     LISTEN      -
tcp        0      0 *:35372                  *:*                     LISTEN      -
tcp        0      0 *:sunrpc                 *:*                     LISTEN      -
tcp        0      0 *:45363                  *:*                     LISTEN      -
tcp        0      0 *:ssh                    *:*                     LISTEN      -
tcp        0      0 localhost:smtp          *:*                     LISTEN      -
tcp        0      0 *:34176                  *:*                     LISTEN      -
tcp6       0      0 [::]:sunrpc             [::]:*                  LISTEN      -
tcp6       0      0 [::]:http                [::]:*                  LISTEN      -
tcp6       0      0 [::]:ftp                 [::]:*                  LISTEN      -
tcp6       0      0 [::]:ssh                 [::]:*                  LISTEN      -
tcp6       0      0 [::]:https               [::]:*                  LISTEN      -
tcp6       0      0 [::]:60477               [::]:*                  LISTEN      -
tcp6       0      0 [::]:46367               [::]:*                  LISTEN      -
tcp6       0      0 [::]:40159               [::]:*                  LISTEN      -
udp        0      0 *:npmp-gui              *:*                     -           -
udp        0      0 *:sunrpc                 *:*                     -           -
udp        0      0 *:699                    *:*                     -           -
udp        0      0 localhost:811            *:*                     -           -
udp        0      0 *:35805                  *:*                     -           -
udp        0      0 *:42502                  *:*                     -           -
udp        0      0 *:npmp-local             *:*                     -           -
udp6       0      0 [::]:sunrpc             [::]:*                  -           -
udp6       0      0 [::]:50309               [::]:*                  -           -
Connected to orb.cs.uow.edu.au  SSH2 - aes128-cbc - hmac-md5 - nc 99x30  NUM
```



- What processes on this machine are listening on the network for incoming connections?
- It's also helpful to look at machines from an outsider's perspective by running a port scanner such as *nmap*.



Examining the routing table

```
orb.cs.uow.edu.au - default - SSH Secure Shell
File Edit View Window Help
Quick Connect Profiles

jyan@orb:~$ netstat -rn
Kernel IP routing table
Destination        Gateway           Genmask          Flags   MSS Window  irtt Iface
0.0.0.0            130.130.64.254   0.0.0.0          UG      0  0        0 eth0
130.130.64.0       0.0.0.0          255.255.255.0    U       0  0        0 eth0
jyan@orb:~$
```

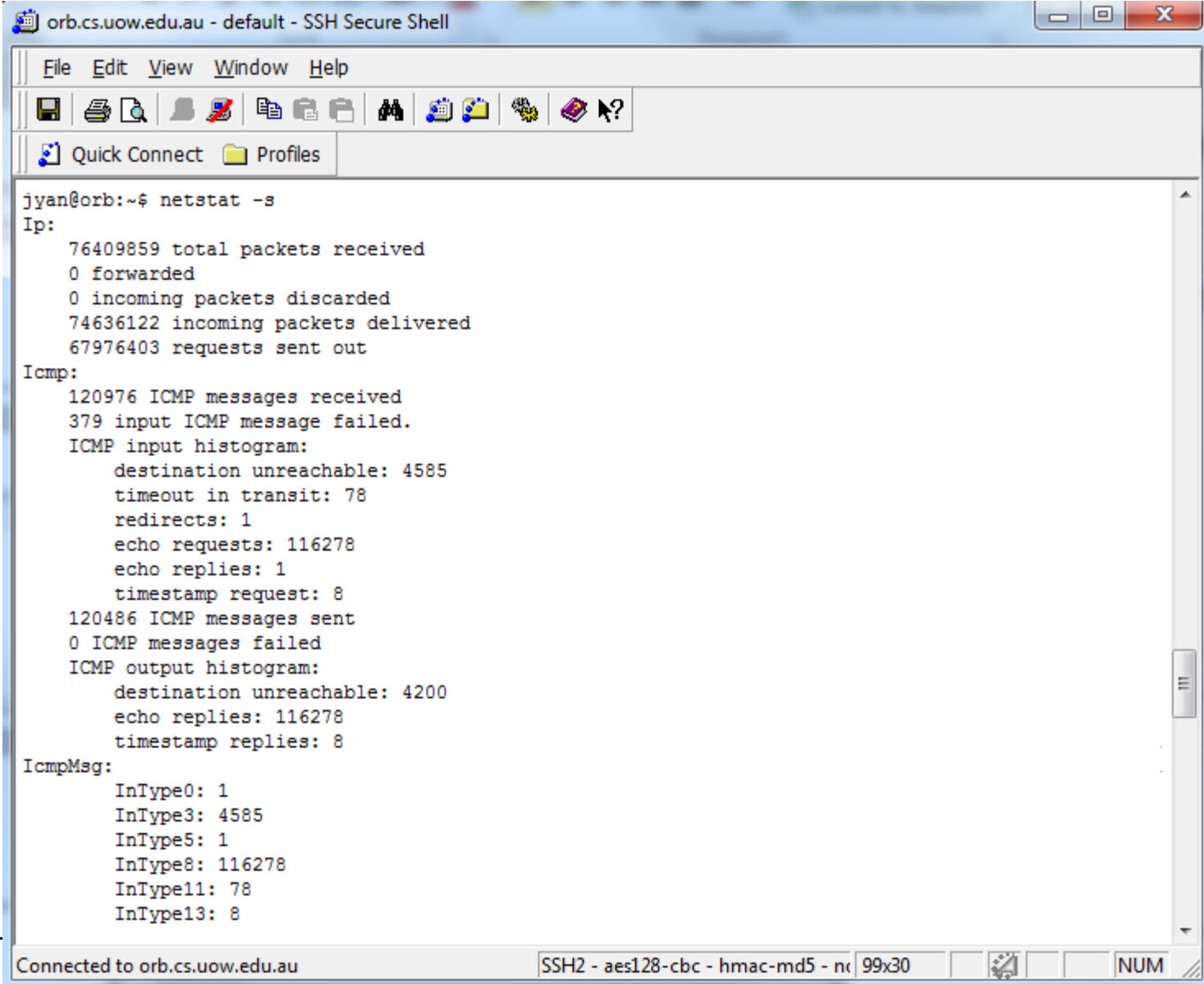
Connected to orb.cs.uow.edu.au SSH2 - aes128-cbc - hmac-md5 - nc 99x30 NUM



- Flags characterise the route
 - U means up
 - G is a gateway
 - H is a host route
 - D indicates a route resulting from an ICMP redirect



Viewing operational statistics



The screenshot shows a terminal window titled "orb.cs.uow.edu.au - default - SSH Secure Shell". The window has a menu bar (File, Edit, View, Window, Help) and a toolbar with various icons. Below the toolbar is a "Quick Connect" button and a "Profiles" folder icon. The terminal content shows the command `netstat -s` being executed, followed by a detailed output of network statistics. The output is organized into sections: "Ip:", "Icmp:", and "IcmpMsg:". The "Ip:" section shows statistics for total packets received, forwarded, discarded, delivered, and sent out. The "Icmp:" section shows statistics for ICMP messages received, failed, sent, and failed, along with an input histogram and an output histogram. The "IcmpMsg:" section shows the distribution of ICMP message types. The status bar at the bottom indicates the connection is to "orb.cs.uow.edu.au" and the encryption is "SSH2 - aes128-cbc - hmac-md5 - n".

```
orb.cs.uow.edu.au - default - SSH Secure Shell
File Edit View Window Help
[Icons]
Quick Connect Profiles

jyan@orb:~$ netstat -s
Ip:
  76409859 total packets received
  0 forwarded
  0 incoming packets discarded
  74636122 incoming packets delivered
  67976403 requests sent out
Icmp:
  120976 ICMP messages received
  379 input ICMP message failed.
  ICMP input histogram:
    destination unreachable: 4585
    timeout in transit: 78
    redirects: 1
    echo requests: 116278
    echo replies: 1
    timestamp request: 8
  120486 ICMP messages sent
  0 ICMP messages failed
  ICMP output histogram:
    destination unreachable: 4200
    echo replies: 116278
    timestamp replies: 8
IcmpMsg:
  InType0: 1
  InType3: 4585
  InType5: 1
  InType8: 116278
  InType11: 78
  InType13: 8

Connected to orb.cs.uow.edu.au  SSH2 - aes128-cbc - hmac-md5 - n  99x30  NUM
```



- *netstat -s* dumps the contents of counters that are scattered throughout the network code.
- The output has separate sections for IP, ICMP, TCP, and UDP
- Pay attention to number of packets that are dropped or discarded
- It's a good idea to develop a feel for the normal ranges of these statistics so that you can recognise pathological states.



Inspection of live interface activity

- One good way to identify network problems is to look at what's happening right now
 - Linux uses a tool called *sar*
 - *sar -n DEV 2 30* shows
 - Number of packets
 - Bandwidth in use
 - Number of compressed packets
 - Number of multicast packets
- Evert 2 seconds for 1 minute



Terminal

jyan@ubuntu: /etc/network

```
jyan@ubuntu:/etc/network$ sar -n DEV 2 30
Linux 3.13.0-115-generic (ubuntu)      03/28/2018      _i686_  (1 CPU)
```

		IFACE	rxpck/s	txpck/s	rxkB/s	txkB/s	rxcmp/s	txcmp/s	rxmcs/s
06:04:59 PM		lo	0.00	0.00	0.00	0.00	0.00	0.00	0.00
06:05:01 PM		eth0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
06:05:01 PM		IFACE	rxpck/s	txpck/s	rxkB/s	txkB/s	rxcmp/s	txcmp/s	rxmcs/s
06:05:03 PM		lo	0.00	0.00	0.00	0.00	0.00	0.00	0.00
06:05:03 PM		eth0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
06:05:03 PM		IFACE	rxpck/s	txpck/s	rxkB/s	txkB/s	rxcmp/s	txcmp/s	rxmcs/s
06:05:05 PM		lo	0.00	0.00	0.00	0.00	0.00	0.00	0.00
06:05:05 PM		eth0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
06:05:05 PM		IFACE	rxpck/s	txpck/s	rxkB/s	txkB/s	rxcmp/s	txcmp/s	rxmcs/s
06:05:07 PM		lo	0.00	0.00	0.00	0.00	0.00	0.00	0.00
06:05:07 PM		eth0	0.00	0.00	0.00	0.00	0.00	0.00	0.00
06:05:07 PM		IFACE	rxpck/s	txpck/s	rxkB/s	txkB/s	rxcmp/s	txcmp/s	rxmcs/s
06:05:09 PM		lo	0.00	0.00	0.00	0.00	0.00	0.00	0.00
06:05:09 PM		eth0	0.00	0.00	0.00	0.00	0.00	0.00	0.00

tcpdump

- *tcpdump* is a packet sniffer that listens to network traffic and record packets that meet criteria of your choice, for example
 - Inspect all packets send to or from a particular host
 - TCP packets related to one particular network connection
- Do not invade the privacy of your users.
- Sniffers must run as root.
- Packet sniffers can produce an overwhelming amount of information.
- You might want to try another tool called Wireshark



Common options

- -i: force an interface
- -n: stop name lookups
- -v: increase the information you see about packets
- -vv: even more data
- -w: store packets to a file (only header)
- -r: read packets from a file



Some examples

\$ sudo tcpdump host bull

\$ sudo tcpdump src net 192.168.1.0/24 and dst port 80



SNMP

- Simple Network Management Protocol
 - Is governed by IETF (RFC 1157)
 - Is designed to be simple
 - Only five commands in version 1
 - Is concerned with network management
 - Manage network performance
 - Identify and diagnose issues
 - Baseline/plan for network growth
 - Key elements
 - Management Station/Manager
 - Agent
 - Management Information Base
 - Network Management Protocol

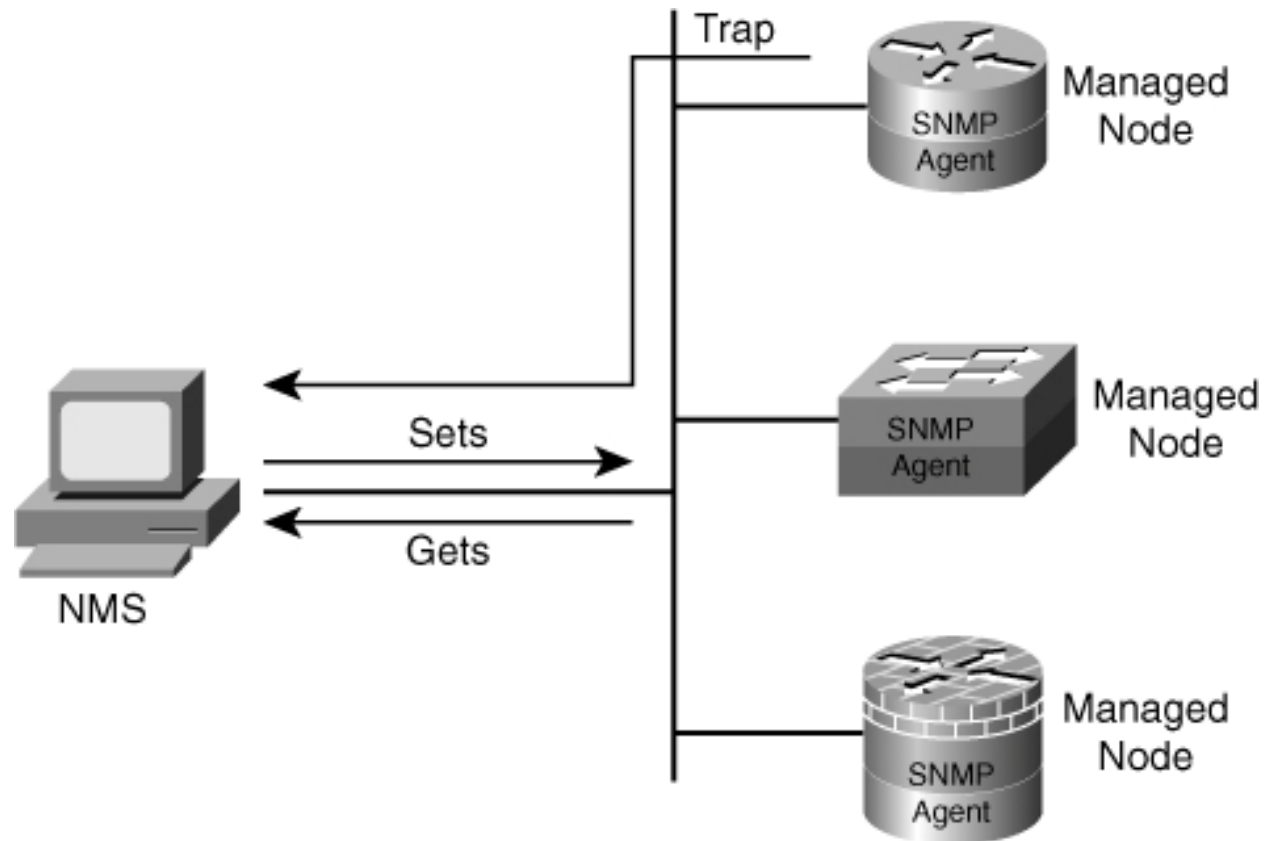


SNMP

- Uses an agent/station model
 - An SNMP agent is software capable of answering valid queries from an SNMP station
 - An SNMP station, such as a network management system, is software capable of forming queries based on information defined in the MIB



SNMP



A client/server architecture



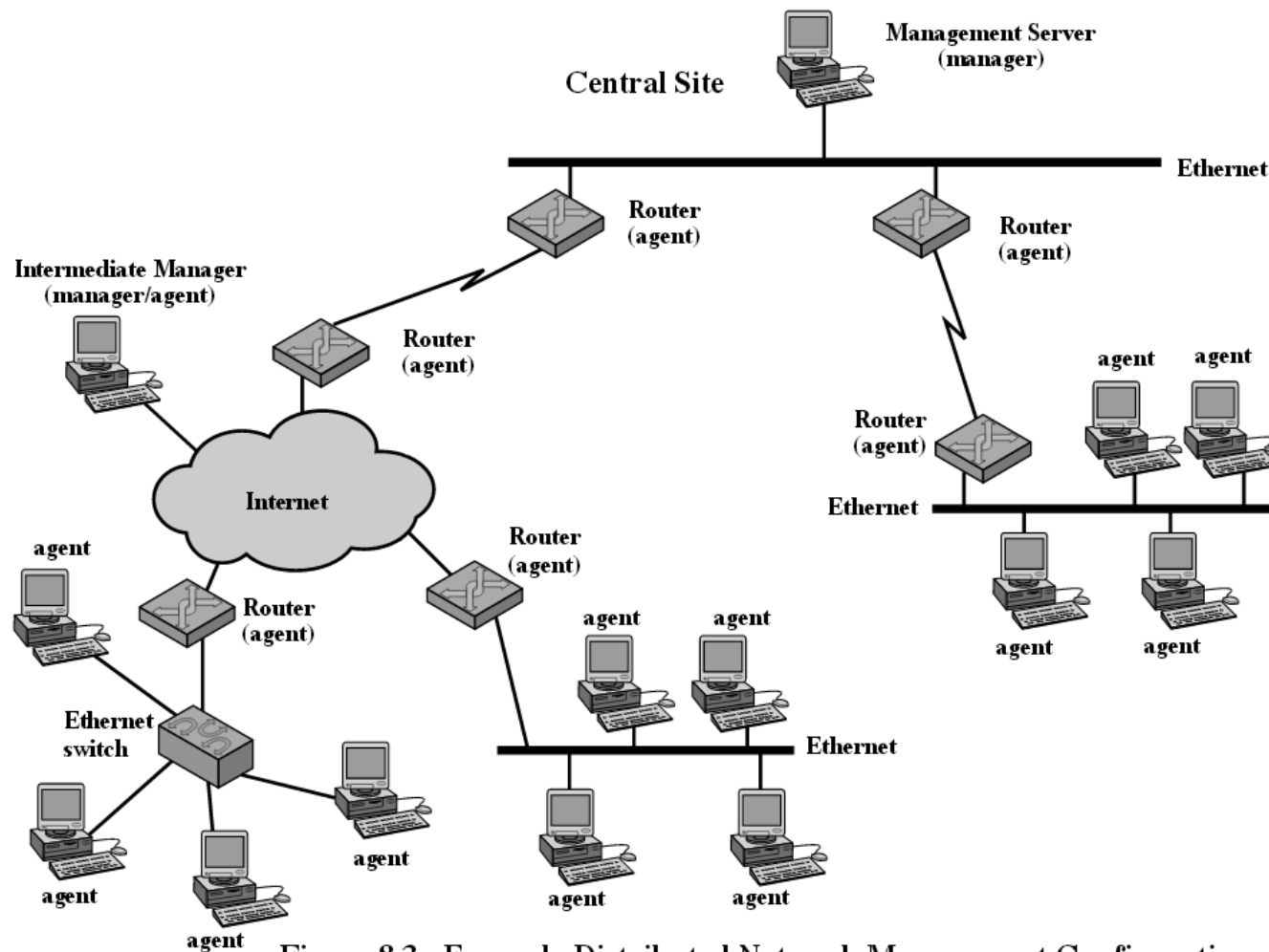
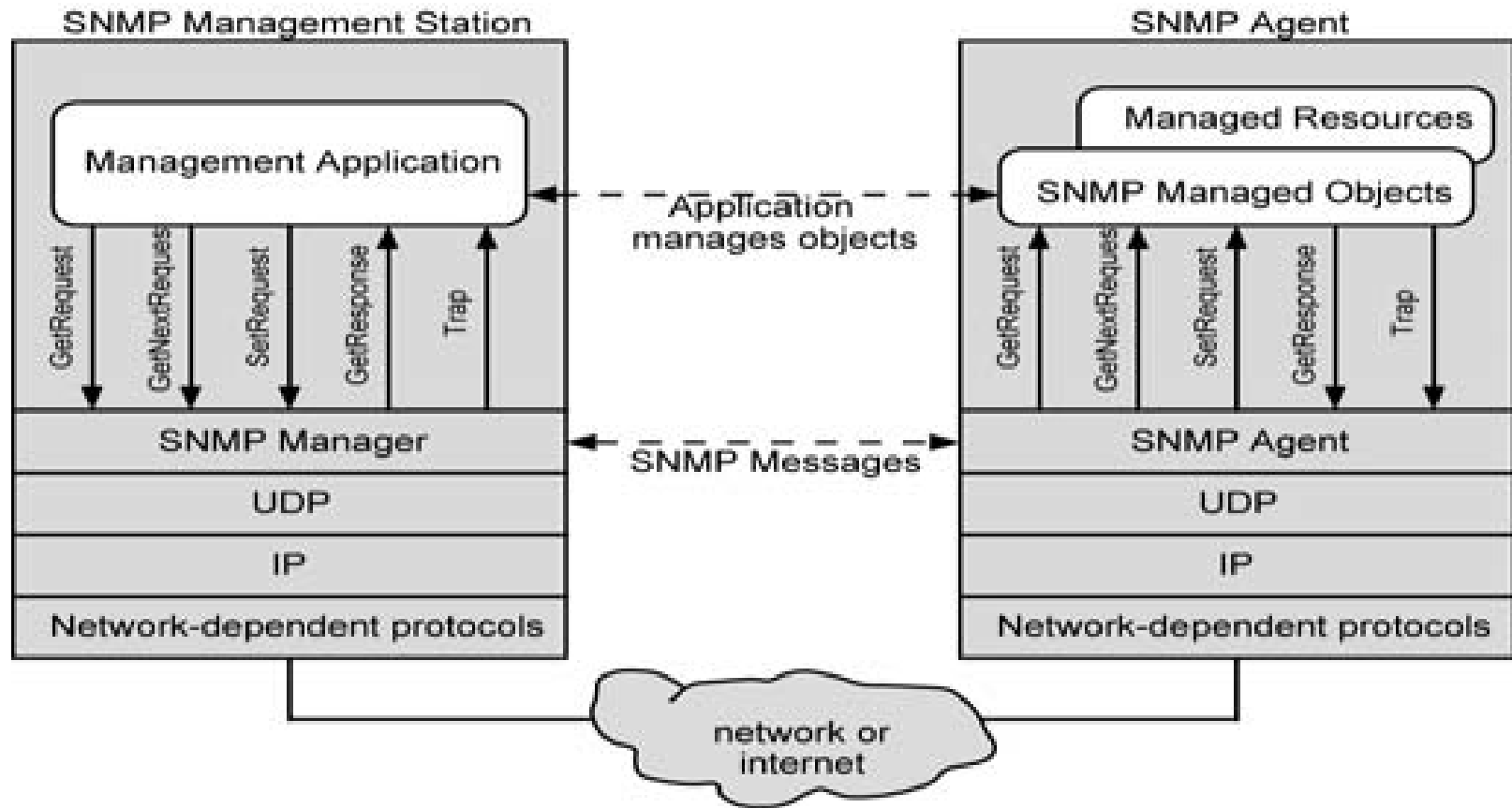


Figure 8.3 Example Distributed Network Management Configuration



SNMP



SNMP configuration



SNMP

- Three basic ideas
 - A manager checks an agent by requesting information that reflects the behaviour of the agent
 - A manager forces an agent to perform a task by resetting values in the agent database
 - An agent contributes to the management process by warning the manager of an unusual situation



SNMP V1

- Get-Request
 - Polls agents for information
- Get-Next-Request
 - Requests the next item in a dataset
- Set
 - Requests a change to a value
- Get-Response
 - Sends a response to a Manager request
- Trap
 - Informs a Manager of an event



SNMP traps

- Seven types of SNMP traps are defined
 - Coldstart of a system
 - Warmstart of a system
 - Link Down
 - Link Up
 - Failure of authentication
 - Exterior Gateway Protocol (EGP) neighbour loss
 - Enterprise specific



SNMP

- Structure of Management Information (SMI)
 - Data definition language for objects
- Management Information Base
 - Distributed network management data



What Is an SNMP-Compliant MIB?

- Each management station or agent in an SNMP-managed network maintains a local database of information relevant to network management, known as the management information base (MIB).
- An MIB contains definitions and information about the properties of managed resources and the services that the agents support. The manageable features of resources, as defined in an SNMP-compliant MIB, are called *managed objects* or *management variables* (or just objects or variables).



What Is an SNMP-Compliant MIB?

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Categories of management information

- State information –
 - this is information about the current state of physical and logical resources along with operational data – provides an instantaneous snapshot in a table
- Physical configuration information
 - how devices are physically configured (device type, physical configuration serial numbers and MAC addresses)



Categories of management information

- Logical configuration information
 - how devices are configured such as IP addresses information; protocols implemented – managed using a CLI (command line interface)
- Historical information
 - Snapshots of performance-related states information (such as packet-counts for each 15 minute interval over the past 24 hours) (usually proprietary format)



Categories of management information

- My observations
 - These categories of information are used and treated differently by management applications
 - Some information can't be set (e.g. serial number of a device)
 - These categories will make more sense after you develop deeper knowledge of the topic



The difference between a MIB and a database

- Footprint: regular DBMS mechanisms are heavier weight and require more processing resources than management interfaces.
- Specific management requirements: DBMSs are not well suited to capturing some of the constraints that are specific and common to management, for example, hierarchical management information.



The difference between a MIB and a database

- Real effects: a MIB is not a “passive” database, but a view on an “active” real-world system. Information in the MIB is accessed through and affected by not only management operations, but many other means as well.
- Characteristics of the contained data: A database typically contains large volumes of data that is largely of the same structure. A MIB contains many different types of information with relatively few instances of each.

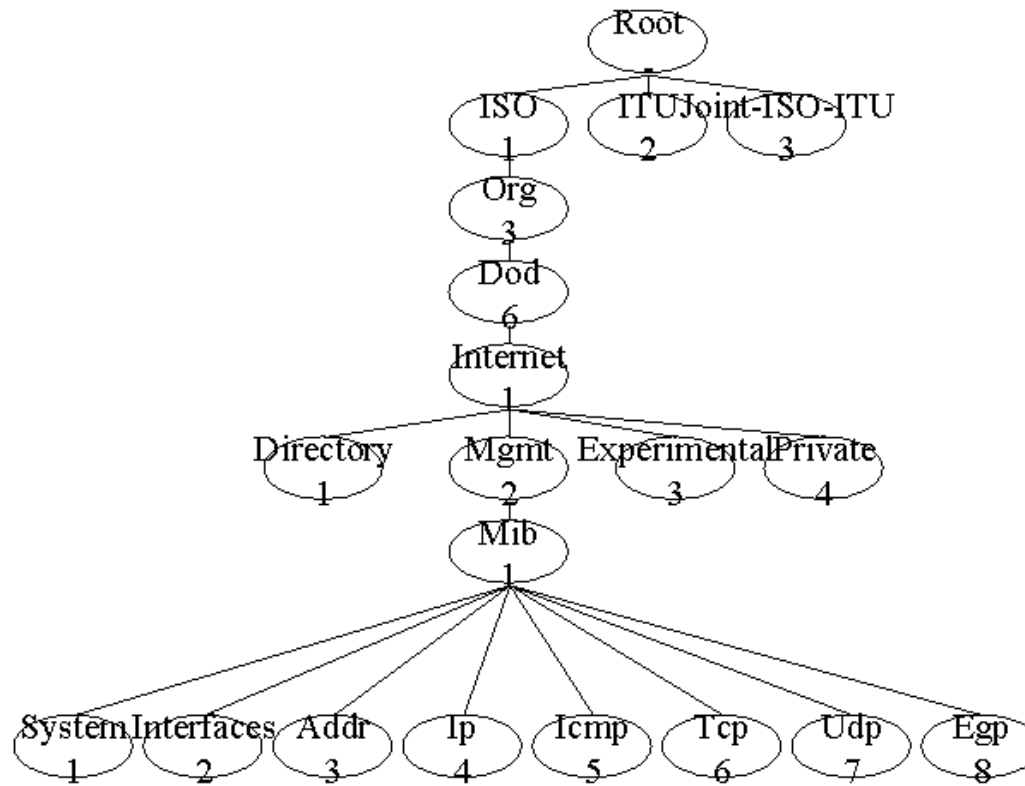


MIB

- All names are globally unique
- All nodes of the name tree are assigned numeric values by standards authorities
 - iso.org.dod.internet.mgmt.mib.ip.ipInReceives has the value 1.3.6.1.2.1.4.3
- All names are specified using a subset of Abstract Syntax Notation (ASN.1)



MIB Namespace



MIB Naming Examples

<u>Object ID</u>	<u>Name</u>	<u>Type</u>	<u>Comments</u>
1.3.6.1.2.1.7.1	UDPInDatagrams	Counter32	total # datagrams delivered at this node
1.3.6.1.2.1.7.2	UDPNoPorts	Counter32	# undeliverable datagrams no app at port
1.3.6.1.2.1.7.3	UDInErrors	Counter32	# undeliverable datagrams all other reasons
1.3.6.1.2.1.7.4	UDPOutDatagrams	Counter32	# datagrams sent
1.3.6.1.2.1.7.5	udpTable	SEQUENCE	one entry for each port in use by app, gives port # and IP address



MIB-II

- An example of a base set of parameters to monitor can be developed from the standard MIB-II. The following parameters can be collected on a per-interface basis:
 - ifInOctetsNumber of bytes received
 - ifOutOctetsNumber of bytes sent
 - ifInUcastPktsNumber of unicast packets received
 - ifOutUcastPktsNumber of unicast packets sent
 - ifInNUcastPktsNumber of multicast/broadcast packets received
 - ifOutNUcastPktsNumber of multicast/broadcast packets sent
 - ifInErrorsNumber of erroneous packets received
 - ifOutErrorsNumber of packets that could not be sent



Limitations and problems with SNMP

- It is officially only standardised for use on IP networks
- It is inefficient at large table retrievals
- It uses cleartext strings for security



SNMP v2

- Same basic function as SNMP
- Original enhancement proposal was concerned primarily with security enhancement
 - Authentication
 - Message integrity
 - Privacy
 - Authorisation and access control



nmap: network port scanner

- nmap checks a set of target hosts to see which TCP and UDP ports have servers listening on them
- Find out what a system looks like to someone on the outside who is trying to break in
- Do not run nmap on someone else's network without permission from that network's administrator



```
Terminal
jyan@ubuntu: /etc
jyan@ubuntu:/etc$ nmap -sT localhost

Starting Nmap 5.21 ( http://nmap.org ) at 2018-03-05 17:08 PST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00013s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE
53/tcp    open  domain
631/tcp   open  ipp

Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
jyan@ubuntu:/etc$
```

Domain Name System (DNS)

- There are hundreds of countries, millions of networks and millions of hosts.
- How do we keep track of all the nodes and networks.
- Well we use a service called DNS (Domain Name Service).
- The Domain Name Service is a *distributed database* which provides a lookup table for *hostnames* and *IP addresses*.
- Humans prefer to refer to nodes on the internet by names, rather than IP addresses. In order for this to work, we need to convert these high level names into addresses. DNS performs name resolution to IP address (and also the opposite direction).
- The Domain Name System also gives us a network hierarchy. It is responsible for giving structure to the Internet (from the users perspective).

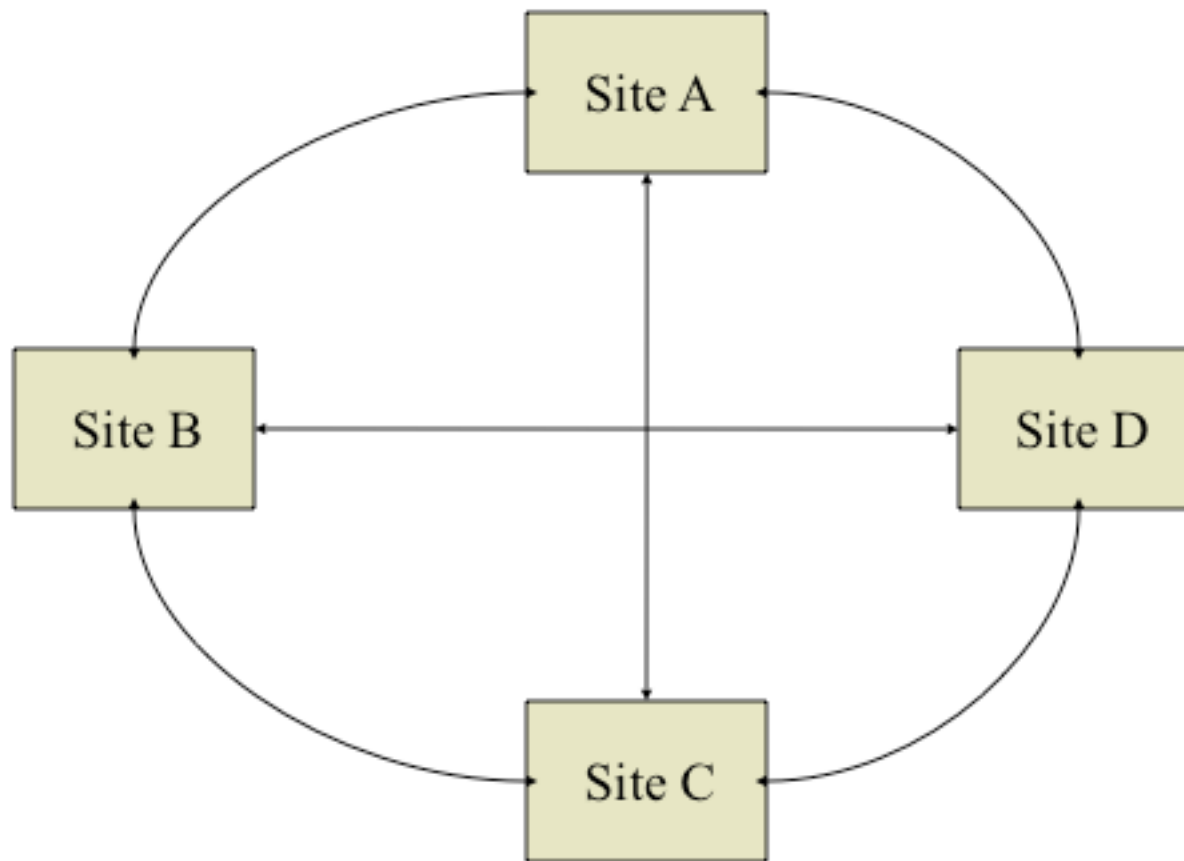


The History Behind the Domain Name Service

- When the Internet was first born (ARPANET)- the mapping between Hostnames and IP addresses was done in small files.
- If a site wished to add a new computer to the network, they choose a name and added it to such files.
- These files were distributed to other sites. Such a process consumed considerable bandwidth.
- That said there were consistent problems keeping the data between sites up to date and in sync.
- It soon became clear that this technique simply did not scale.
- This mapping between names and IP addresses is extremely important because as humans we prefer to remember names as opposed to long sequences of numbers.
- On Unix, you have already seen how the `/etc/hosts` file may contain such mappings;

130.130.217.198 wumpus.its.uow.edu.au





- The exchange of host files between sites was too expensive, complex and inefficient.



The History Behind the Domain Name Service

- The Domain Name System was created to address two problems;
 - The organization of the internet.

As ARPANET grew researchers began to realize that names for hosts were running out. So they needed to impose a hierarchy/structure.

- The synchronization of host databases.
Again as ARPANET grew it became more and more difficult to keep track of nodes. Data quickly became dated.



The History Behind the Domain Name Service

- DNS was formally introduced by Paul Mockapetris in RFC's 882 and 883 (around 1983).
- The original version of DNS was designed and implemented at the University of California at Berkeley in 1984.
- In 1985 the Berkeley Computer Systems Research Group absorbed the graduate students work and rolled it into the BSD project (Berkeley Software Distribution)
- Kevin Dunlap in 1985 released BIND (Berkeley Internet Name Domain system) which has grown in popularity over the years.
- Bind is currently in release 9 and an organisation called the **Internet Software Consortium** (ISC) has evolved from the work done at Berkeley. The ISC is a non-profit organisation funded by the USA government, community and private sector.



The History Behind the Domain Name Service

- DNS has evolved since those early days and continues to evolve. Initiatives include:
 - DNSSEC
Secure DNS transactions i.e. the ability to move data between servers in a secure and trusted fashion.
 - IPv6 Support
With the adoption of IPv6, DNS has had to change only slightly.
- It has also been an interesting time because vendors such as Microsoft have come up with several good ideas which have been rolled back into the DNS system.
 - For example Microsoft has developed the notion of Dynamic DNS which has had a profound impact.



So What *is* DNS?

- In order to understand DNS, you need to understand the Internet.
- The internet is made up of sites. These sites reflect networks or organisations.
- Organisations can be grouped together depending on type.
- Entities can be grouped together based upon geographic locality.
- Ultimately DNS provides the following kind of functionality;
 - A hierarchal namespace for IP addresses.
 - A distributed database of host information.
 - A mechanism for finding services and nodes on a network.
 - A protocol for exchanging naming information to keep data current and synchronised between sites.



The Domain Name System Architecture

- But how does it work?
 - The Domain name system defines a tree of domains. Each domain represents a chunk or grouping of the network which is managed by a single administrative entity.
 - The Domain Name Service is a global infrastructure managed by ICANN.
 - The Domain Name System defines what are known as Top Level Domains (TLDs).
 - These TLDs had not been changed in years. It used to be that certain TLDs represented specific kinds of networks.
 - But as the Internet has grown, ICANN added a few top level domains. And more recently, hundreds more!



gTLD

- Top level domains are sometimes referred to in the literature are gTLD's (generic top level domains).

Domain	Use	Domain	Use
com	Commercial Entities	net	Network Providers
edu	Education Entities	org	Non profit organisations
gov	Government	int	International
mil	Military Networks	arpa	Old Style Arpanet



ccTLD

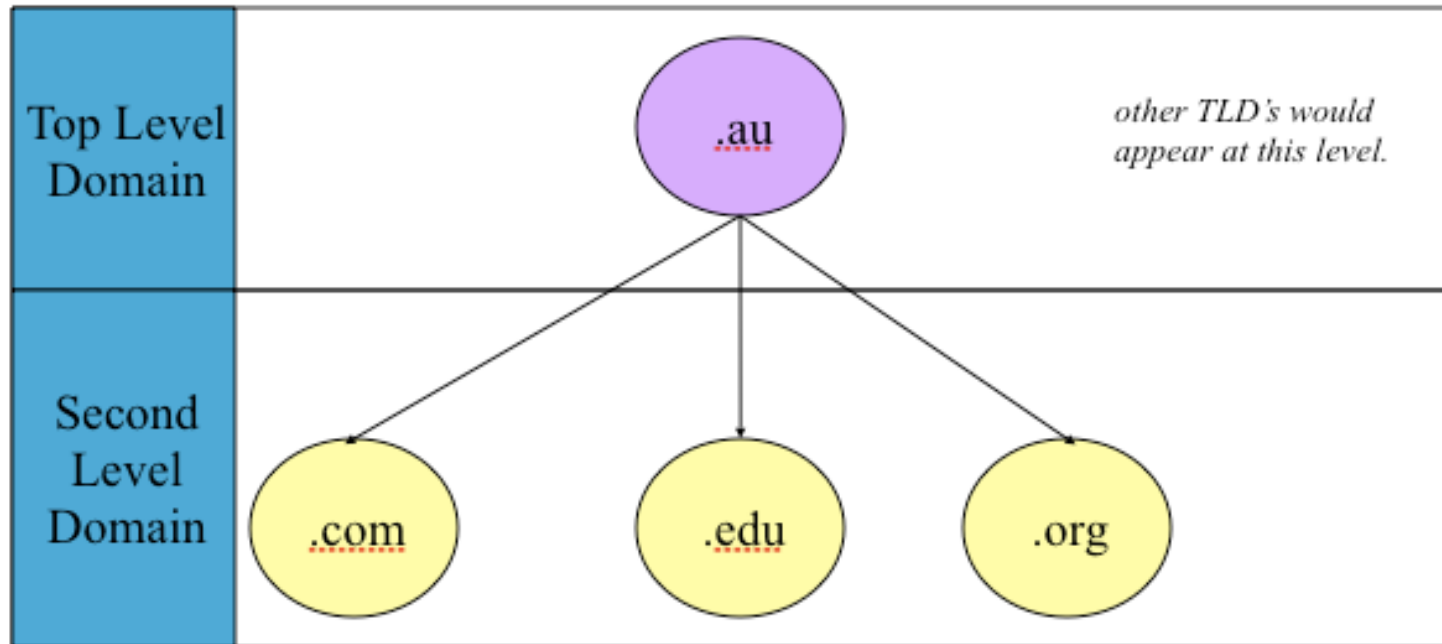
- Most of the top level domains are used within the United States. Outside the United states there are a number of top level country code domains *ccTLD*'s. Some common country codes are

Code	County	Code	Country	Code	Country
au	Australia	fi	Finland	hk	Hong Kong
ca	Canada	fr	France	ch	Switzerland
br	Brazil	jp	Japan	mx	Mexico
de	Germany	se	Sweden	mu	Hungary

- Most of these countries produce second level domains which are similar to the top level domains offered by the US.



- For example the internet has the following structure.



Subdomain

- You can infer from the DNS model that domains can be delegated to other authorities (sub domains).
- **.au** has its own authority
- And at the University, **EIS** have their own subdomain beneath the **uow** domain. The management of this **eis.uow.edu.au** domain is partially delegated.



Regional domains

- Domains largely emerge due to congestion in the namespace - think about how many movies have been put in the **.com** domain!
- Regional top level domains are managed by local authorities. For example in Australia **.au** domain registry is performed by *AusRegistry*, overseen by industry regulator **auDA** (**.au Domain Administration**). The control is authorised by ICANN.
- The management of educational domains in Australia is done by a single private sector organization (Educause <http://www.educause.edu/edudomain/>)
- If an organisation wishes to have a **.com.au** or **.edu.au** domain they need to speak to the appropriate authority.
- Each domain on the Internet has at least one DNS server which keeps track of the network.
- These DNS servers talk to the parent server and serve requests of clients around the world.
- The top level domains have what are known as ‘**root**’ name servers. These process a huge number of requests and updates.



Obtaining a Domain.

- Depending upon where you wish to create a domain you need to work with different agencies.
- ICANN is responsible for delegating ‘authority’ to agencies to handle certain top level domains.
- In Asia Pacific, `www.apnic.net` has a registry of all the domain and organisations that are managed.



Technical Details of DNS

- There is no implicit tie between network numbers and domain names
- However, it makes administrative sense to keep some relationship between networks and names.
- The University of Wollongong domain happens to map nicely onto our network (although we have some smaller ranges in there as well).
- 130.130 – uow.edu.au
- Because we are responsible for the domain we run a DNS server such as bind.
- The people in the domain above us (edu.au) have told their DNS server that our DNS server is authoritative for the uow.edu.au sub domain.
- The domain uow.edu.au is a delegation point in the edu.au domain.



- Small organisations can run servers on their own hosts or ask their ISP to supply DNS service
- Medium-sized sites should run multiple DNS servers to reduce query latency
- A very large site can divide its DNS domain into subdomains and run several server for each subdomain



Technical Details of DNS

- Inside `uow.edu.au` there is a sub domain `cs.uow.edu.au` (plus many others).
- Although it is a sub domain, it is not a delegation point since the `uow.edu.au` DNS server is authoritative for that domain.
- A DNS server may be authoritative for sub domains or may delegate responsibility to other DNS servers.



Types of DNS

- There are several kinds of DNS Server
 - Authoritative

An authoritative server is one that is authorised to answer queries for a domain. It is guaranteed to be up to date. It is considered the official representative of a zone/domain.

- Non-Authoritative.
A non-authoritative server answers a query typically from a cache. It does not know if the data is still valid.
- In the Authoritative grouping we typically have these kinds of servers
 - Masters (Primary)
 - Slaves (Secondary)
- The slave gets a copy of its records from the master using the SOA (Start of Authority) record for the zone.
- Data is maintained between the master and slave using Zone Transfers (over TCP). We can control zone transfers using software e.g. ndc.
- Non-Authoritative servers typically include things such as caches and forwarders.



Recursive vs Non Recursive Queries

- There are two ways in which DNS servers can answer queries
 - Non-Recursive
If a DNS server is non-recursive it will only answer queries that it has the answer for. These may be cached or they may be authoritative.

If it does not have the answer it returns the address of the server that is likely to know the answer.
 - Recursive
A recursive DNS server returns only names or errors. It follows the referrals to yield the answer.

The way this is done is by forwarding the query to a *root* server that is known.



Zone

- A zone is essentially a domain minus its subdomains
- Each zone typically has one master name server that keeps the official copy of the zone's data on disk
- A zone can have several slave name servers that get their data from the master server through a zone transfer operation
- Typically we have two zones;
 - Forward zone name to address mapping.
 - Reverse zone address to name mapping.
- You may hear things like caching-only name server, non-authoritative name server, recursive name server, etc.



DNS Records

- The information in the DNS system consists of records
 - SOA Start of Authority
 - NS Identifies Zones Servers
 - A Name to Address Mapping
 - PTR Address to Name Mapping
 - MX Mail Exchange (used for mail routing)
 - CNAME Synonyms for hosts i.e. Nicknames
 - Glue A Record for a Nameserver



SOA

- The SOA record indicates the beginning of a Zone.
- In many regards a zone can be considered a grouping in the DNS namespace.
- Typically we have two zones
 - Forward zone for name to address mapping.
 - Reverse zone for address to name mapping.
- The SOA record typically contains the following
 - The name of the Zone.
 - The administrator for the Zone.
 - Timeout information e.g.
 - `Serial` is typically used to indicate last change.
 - `Refresh` specifies how often slave servers check with the master to see if the serial has changed.
 - `Retry` specifies how long after a failure a slave checks the master.
- The SOA information is used by slaves to identify when it is necessary to do a Zone Transfer.
- A Zone Transfer allows master and slave DNS servers to remain in sync.



NS

- The NS record identifies which servers are authoritative for a zone. This is how we define master and secondary DNS servers.
- If we do the following we get some interesting results:

```
$ dig uow.edu.au NS
```

- Each domain should have a primary server and at least one secondary.
- Primary and secondary name servers talk among themselves to ensure the consistency of the database.



A

- The A records provide the name to address mappings in the DNS database. They basically represent the `/etc/hosts` file.
- An A record would typically look like this:

```
yoshi IN A 130.130.64.68
```

This entry would be in the zone file representing the `cs` subdomain for the University.



PTR

- The PTR records do the reverse of the A records.
- A reverse DNS lookup for an IP address is NOT a search of the A records in reverse; it is a search of the PTR records for the IP address.
- PTR records exist in a special part of the DNS namespace called *in-addr.arpa*.



In-addr.arpa

- Domains under `in-addr.arpa` are named liked IP addresses with their bytes reversed.

For example at the University the `cs` zone is (was)

`130.130.64.0/24`

Thus the `in-addr.arpa` file will be called `64.130.130.in-addr.arpa`.

- It will typically contain records such as:

`68 PTR yoshi`

- Some software requires names to be reversed for security purposes e.g. `Telnet` or `SSH`.



MX

- MX records are created to help routing of email.
- A MX record knows where mail should be routed.
- For a domain we can have more than one MX record. The mail router uses the MX record with the lowest priority that is available.



MX Example

– If I run:

```
$ dig uow.edu.au MX
```

```
;; ANSWER SECTION:
```

```
uow.edu.au. 3600 IN MX 100 smtp-gateway.uow.edu.au.
```

I will get the MX record for the domain which points to the host smtp-gateway.its.uow.edu.au.

It has a priority of 100.

– If I do the same thing at gmail

```
$ dig gmail.com MX
```

```
;; ANSWER SECTION:
```

```
gmail.com. 455 IN MX 10 alt1.gmail-smtp-in.l.google.com.
```

```
gmail.com. 455 IN MX 5 gmail-smtp-in.l.google.com.
```

```
gmail.com. 455 IN MX 40 alt4.gmail-smtp-in.l.google.com.
```

```
gmail.com. 455 IN MX 30 alt3.gmail-smtp-in.l.google.com.
```

```
gmail.com. 455 IN MX 20 alt2.gmail-smtp-in.l.google.com.
```

we see five hosts. The priority decides where mail goes. We use this for redundancy purposes.



CNAME

- CNAMEs describe canonical names for hosts in the DNS namespace.

In the namespace we will typically see things like this:

wyrm	IN	A	130.130.68.3
smtp	IN	CNAME	wyrm
ftp	IN	CNAME	wyrm



Other records

- There are a few other records which are important to mention:
 - SRV records
SRV records describe priority, ports and other information for network resources/ hosts. This is particularly important in Zero Configuration networking.
 - TXT records
A TXT record adds arbitrary text to a hosts DNS record.



Adding a new machine to DNS (Bind server)

1. Assign an unused name and address to the new machine
2. Log into the master names server (use *dig* to find it)
3. Find the name server configuration file (usually /etc/named.conf). Find the zone data files in the zone statement, e.g.,

```
zone "example.com" {  
    type master;  
    file "filename";  
zone "188.77.208.in-addr.arpa" {  
    type master;  
    file "filename";
```



4. Identify the record of a similar machine in the same subnet, e.g.,

```
template      IN A    208.77.188.100  
              IN MX 10 mail-hub
```

5. Duplicate this record and change it accordingly
6. Edit the reverse zone file

```
100          IN PTR  template.example.com.
```

7. Change the serial number in the SOA record at the beginning of the file
8. Reload the domain

```
sudo rndc reload forward-zone-filename  
sudo rndc reload reverse-zone-filename
```



Hosts on the Network.

- Machines may have many IP addresses, either on different hardware interfaces or the same hardware interface.
- Each IP address may have its own name or not.
- The name of the machine (uname) may not match any interface name.
- Sensible people adopt a sensible naming scheme.
- For example,
 - Machine name matches DNS name for main interface.
 - Each IP address has its own name.
 - Each interface has a single IP address and name.
 - Secondary interface names are derived from the primary name e.g. wraith-1



The Importance of DNS

- Management of their IP addresses and DNS configuration is a complex and onerous task for an organisation of any size.
- Do not underestimate the size of this problem or the complexity of DNS configuration.
- Remember that nearly all network services depend on name resolution.
- ***If DNS fails, most of your services will be down.***
- After your network fabric, DNS is probably the most important network service you will ever have.



The Resolver.

- The resolver on a host is responsible for converting names to addresses. It forwards queries to the appropriate DNS servers.
- Name server configuration is typically via a single configuration file. For Unix, that configuration file is called `/etc/resolv.conf`. (not a typo!)
- Such a file looks like this:

```
domain  its.uow.edu.au
nameserver 130.130.68.1
nameserver 130.130.64.1
nameserver 129.78.64.1
options retry:2 retrans:5
search uow.edu.au cs.uow.edu.au
```



resolv.conf example

- The file has a number of directives. The first is the Primary DNS domain for the current host.
 - `domain its.uow.edu.au`
- The next group of directives specify the name server to use to resolve queries. Queries are sent to each server in order.
 - `nameserver 130.130.68.1`
 - `nameserver 130.130.64.1`
 - `nameserver 129.78.64.1`
- For each server we can specify a series of option. In this case we are saying:
 - `options retry:2 retrans:5`
 - Wait 5 seconds for a reply.
 - Retry each nameserver 2 times before moving on and trying the next.
- This is a very aggressive configuration.
- If name does not resolve, try these domains as well.
 - `search uow.edu.au cs.uow.edu.au`
- This is typically used when we do not provide a fully qualified DNS name.



Multi-homed Hosts

- Machines may have multiple interfaces. We often see these devices in the device tree.
 - `/dev/le0`
 - `/dev/hme4`
- Individual interfaces may be up or down.
- The `ifconfig` command is used to display information about or configure an interface.
- Information displayed includes the hardware address, the IP address, the broadcast address, the netmask, the MTU and some flags.



Important Files

- Under Solaris, `/etc/hostname.X` controls the name (and therefore the IP address) of each interface.
 - `/etc/nodename` contains the nodename or system name of the machine.
 - Name/IP information comes from `/etc/hosts` (needed since before interface operational, no way to get information from DNS)
 - Netmask information from `/etc/netmasks`
 - Network/name translation from `/etc/networks`
- As a systems administrator you need to remember these files.
 - On most Unix implementations they are the same.

